Strategic Product Placement Analysis – Documentation (Flask + Tableau)

📌 **Project Title:** Strategic Product Placement Analysis
Project Type: Modern Data Analytics Website
Tech Stack: Flask, HTML/CSS/JS, Tableau Public (Embedded)

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Abstract

Strategic Product Placement Analysis is a modern analytics website that shows insights about how products are placed in content and how those placements perform. The system collects placement data, prepares it, and displays results through interactive Tableau dashboards and stories embedded inside a website built using Flask.

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1. Project Overview

1.1 Problem Statement

Product placement decisions are often made using manual tracking and static reports. This causes:

- Slow analysis
- Difficulty comparing brands and content
- Lack of interactive filtering
- Poor visibility into trends

1.2 Proposed Solution

A web platform that:

- Displays interactive dashboards for placement analysis
- Allows users to view trends, comparisons, and summaries
- Embeds Tableau dashboards and stories into a clean UI

1.3 Scope of the Project

Included:

- Flask-based website
- Responsive single-page style UI (multiple sections)
- Tableau Public dashboard embed
- Tableau Public story embed
- Contact section (static form)

Not included (can be future work):

- Real-time data ingestion
- User authentication and admin panel
- Database storage and full CRUD

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2. Objectives

2.1 Main Objectives

- Build a professional analytics website for product placement insights.
- Embed Tableau dashboard and story for interactive visualization.
- Provide a clean UI with separate sections (Home, About, Dashboard, Story, Contact).

2.2 Specific Objectives

- Show placement performance in a simple and understandable way.
- Provide a responsive design for different screen sizes.

- Keep navigation smooth using fixed navbar and scroll sections.

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3. Technologies Used

3.1 Backend

- **Python (Flask)**
  - Routing and template rendering
  - Connecting HTML pages using `render_template()`

3.2 Frontend

- **HTML5** for page structure
- **CSS3** for styling (modern dark theme)
- **JavaScript** for Tableau embed script setup

3.3 Visualization Tool

- **Tableau Public**
  - Dashboard embed
  - Story embed

3.4 Tools

- VS Code / PyCharm (IDE)
- Browser (testing)
- Git (optional version control)

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4. Website Sections Explanation

The website is built as a single scrollable page with multiple sections:

4.1 Home (Hero Section)

Purpose:

- Introduce the project name clearly
- Give a short tagline

UI elements:

- Full-screen background image (hero banner)
- Large heading and short description

4.2 About

Purpose:

- Explain what the project involves

Content shown:

- Data collection
- Data preparation
- Tableau visualization

4.3 Dashboard

Purpose:

- Show the **Product Placement Analysis Dashboard**

How it works:

- An embedded Tableau Public dashboard appears inside a styled container.
- Users can interact with filters directly in Tableau.

4.4 Story

Purpose:

- Show the **Product Placement Analysis Story**

How it works:

- An embedded Tableau Public story is displayed for step-by-step narrative insights.

4.5 Contact

Purpose:

- Provide basic contact details and a contact form UI

Includes:

- Location, email, phone (static)
- Input fields (name, email, subject, message)

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5. Dashboard Description

The dashboard is designed to help users understand placements quickly.

5.1 Dashboard Goals

- Summarize key placement results
- Make comparisons easier
- Provide interactive exploration

5.2 Typical Dashboard KPIs (Example)

- Total placements
- Placements by brand
- Placements by platform/content type
- Trend over time
- Engagement or reach (if data is available)

5.3 Interactions

- Filter by category or time (Tableau filters)

- Hover tooltips for details
- Click on charts to highlight related values

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6. UI Design Features

6.1 Design Style

- Dark theme background
- Modern typography (Segoe UI)
- Glass-effect navbar (`backdrop-filter: blur()`)

6.2 Navigation

- Fixed top navbar
- Smooth scrolling to sections using anchor links

6.3 Layout and Components

- Reusable spacing using `.section`
- Content arranged using flexbox in About and Contact sections
- Card-style container for Tableau embeds (`.tableau-box`)

6.4 User Experience

- Hover effects on links and cards
- Clear headings and spacing
- Simple and readable content

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7. Methodology

7.1 Requirement Analysis

- Identify sections needed: Home, About, Dashboard, Story, Contact

- Finalize what should be shown in dashboard and story
- Choose Flask for simple backend routing

7.2 Design

- Create a basic UI wireframe (navbar + sections)
- Decide a consistent color palette and fonts

7.3 Implementation

- Implement Flask routing
- Build HTML template and CSS styling
- Add Tableau embed placeholders
- Add JavaScript to load Tableau embed script

7.4 Testing

- Check loading of all sections
- Verify responsive behavior
- Validate that dashboard and story load correctly
- Verify smooth scrolling and navbar links

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8. Conclusion

Strategic Product Placement Analysis successfully provides a modern analytics website structure using Flask and Tableau. It presents interactive dashboards and stories with a professional UI. This makes product placement insights easier to understand and more useful for decision-making.

8.1 Future Enhancements

- Add login and role-based access
- Add database integration (store placement data)

- Add automated data upload and preprocessing
- Add more dashboards and downloadable reports

Appendix A: Source Code

A1. `app.py` (Flask)

```
from flask import Flask, render_template

app = Flask(__name__)

@app.route("/")
def home():
    return render_template("index.html")

if __name__ == "__main__":
    app.run(debug=True)
```

A2. `templates/index.html`

```
<!DOCTYPE html>
<html>
<head>
    <title>Strategic Product Placement Analysis</title>
    <style>
        *{margin:0;padding:0;box-sizing:border-box;}
        body{font-family:Segoe UI,Arial;background:#070707;co
lor:white;scroll-behavior:smooth;}

        /* ===== NAVBAR ===== */
        nav{position:fixed;top:0;width:100%;display:flex;alig
n-items:center;justify-content:space-between;padding:18px 70p
x;background:rgba(0,0,0,0.65);backdrop-filter:blur(15px);z-in
```

```
dex:1000;border-bottom:1px solid rgba(255,255,255,0.08);}
        .logo{font-weight:700;letter-spacing:2px;}
        .nav-links{position:absolute;left:50%;transform:trans
lateX(-50%);}
        .nav-links a{color:white;margin:0 18px;text-decoratio
n:none;font-weight:600;transition:0.3s;}
        .nav-links a:hover{color:#f5b041;}
        .btn{border:1px solid #f5b041;padding:8px 18px;border
-radius:8px;cursor:pointer;}

        /* ===== SECTION ===== */
        .section{padding:130px 90px;max-width:1400px;margin:a
uto;}
        h2{font-size:34px;margin-bottom:25px;}

        /* ===== HERO ===== */
        .hero{height:100vh;background:linear-gradient(rgba(0,
0,0,0.6),rgba(0,0,0,0.85)),url("https://images.unsplash.com/p
hoto-1556742031-c6961e8560b0")center/cover no-repeat;display:
flex;justify-content:center;align-items:center;text-align:cen
ter;}
        .hero h1{font-size:58px;margin-bottom:10px;}
        .hero p{color:#ddd;font-size:18px;}

        /* ===== ABOUT ===== */
        .about{display:flex;gap:60px;flex-wrap:wrap;align-ite
ms:center;}
        .about img{width:520px;border-radius:16px;box-shadow:
0 20px 40px rgba(0,0,0,0.6);transition:0.4s;}
        .about img:hover{transform:scale(1.03);}

        /* ===== TABLEAU BOX ===== */
        .tableau-box{background:#fff;padding:28px;border-radi
us:16px;box-shadow:0 10px 30px rgba(0,0,0,0.5);color:black;tr
ansition:0.4s;}
        .tableau-box:hover{transform:translateY(-4px);}
```

```
        /* ===== CONTACT ===== */
        .contact-box{display:flex;gap:40px;flex-wrap:wrap;}
        .info{background:#111;padding:30px;width:320px;border
-radius:12px;}
        .form{background:#f2f2f2;color:black;padding:30px;fle
x:1;border-radius:12px;}
        input,textarea{width:100%;padding:12px;margin-top:12p
x;border-radius:6px;border:none;font-size:15px;}
        button{background:#f5b041;border:none;padding:12px 28
px;margin-top:15px;cursor:pointer;border-radius:6px;font-weig
ht:bold;transition:0.3s;}
        button:hover{background:#ffcc66;transform:translateY
(-2px);}
    </style>
</head>

<body>
    <!-- NAVBAR -->
    <nav>
        <h2 class="logo">SALES</h2>
        <div class="nav-links">
            <a href="#home">Home</a>
            <a href="#about">About</a>
            <a href="#dashboard">Dashboard</a>
            <a href="#story">Story</a>
            <a href="#contact">Contact</a>
        </div>
        <a class="btn">Get Started</a>
    </nav>

    <!-- HERO -->
    <section id="home" class="hero">
        <div>
            <h1>Strategic Product Placement Analysis</h1>
            <p>We are a team of talented Data Analytics</p>
```

```
        </div>
    </section>

    <!-- ABOUT -->
    <section id="about" class="section about">
        <div>
            <h2>The Project Involves</h2>
            <p>✔ Data collection gathers business insights.</
p>
            <p>✔ Data preparation transforms raw data.</p>
            <p>✔ Tableau visualization shows trends clearly.
</p>
        </div>
        <img src="https://images.unsplash.com/photo-154283813
2-92c53300491e">
    </section>

    <!-- DASHBOARD -->
    <section id="dashboard" class="section">
        <h2>Product Placement Analysis Dashboard</h2>
        <div class="tableau-box">
            <div class='tableauPlaceholder' id='vizDashboar
d'>
                <object class='tableauViz'>
                    <param name='host_url' value='https://pub
lic.tableau.com/' />
                    <param name='embed_code_version' value
='3' />
                    <param name='path' value='shared/PGDZFZH5
3' />
                    <param name='toolbar' value='yes' />
                </object>
            </div>
        </div>
    </section>
```

```
    <!-- STORY -->
    <section id="story" class="section">
        <h2>Product Placement Analysis Story</h2>
        <div class="tableau-box">
            <div class='tableauPlaceholder' id='vizStory'>
                <object class='tableauViz'>
                    <param name='host_url' value='https://pub
lic.tableau.com/' />
                    <param name='embed_code_version' value
='3' />
                    <param name='path' value='shared/2TCS424K
3' />
                    <param name='toolbar' value='yes' />
                </object>
            </div>
        </div>
    </section>

    <!-- CONTACT -->
    <section id="contact" class="section">
        <h2>CONTACT US</h2>
        <div class="contact-box">
            <div class="info">
                <p><b>📍 Location:</b><br>Gachibowli Circle,
Hyderabad, INDIA</p>
                <p><b>✉ Email:</b><br>businessstore@example.c
om</p>
                <p><b>📞 Call:</b><br>89554 48855</p>
            </div>
            <div class="form">
                <input type="text" placeholder="Your Name">
                <input type="email" placeholder="Your Email">
                <input type="text" placeholder="Subject">
                <textarea placeholder="Message"></textarea>
                <button>Send Message</button>
            </div>
```

```
        </div>
    </section>

    <script>
        var divElement = document.getElementById('vizDashboar
d');
        var vizElement = divElement.getElementsByTagName('obj
ect')[0];
        vizElement.style.width = '100%';
        vizElement.style.height = '800px';

        var scriptElement = document.createElement('script');
        scriptElement.src = 'https://public.tableau.com/javas
cripts/api/viz_v1.js';
        vizElement.parentNode.insertBefore(scriptElement, viz
Element);

        var divElement2 = document.getElementById('vizStor
y');
        var vizElement2 = divElement2.getElementsByTagName('o
bject')[0];
        vizElement2.style.width = '100%';
        vizElement2.style.height = '900px';
    </script>
</body>
</html>
```